INCH-POUND

MIL-W-71094 (AR)
24 June 1992

MILITARY SPECIFICATION

WAXES, COAL-BASED, FOR USE IN R8150 EXPLOSIVE

This specification is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality acceptance provisions for coal-based waxes, Type I and Type II (see 6.1.1) for the manufacture of explosive compositions.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provision, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9160
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STANDARDS

MILITARY

MIL-STD-650 - Explosive: Sampling, Inspection and Testing.
MIL-STD-1168 - Lot Numbering of Ammunition

(Unless otherwise indicated, copies of the federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D92 - Test for Flash and Fire Points by Cleveland Open Cup

D127 - Test for Drop Melting Point of Petroleum Wax

D445 - Test for Kinematic Viscosity of Transparent and Opaque Liquids

D721 - Test for Oil Content of Petroleum Waxes,

D938 - Congealing Point of Petroleum Waxes, Including Petrolatum

D1321 - Test for Needle Penetration of Petroleum Waxes

D3951 - Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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AMERICAN OIL CHEMISTS SOCIETY (A.O.C.S.)

Cc 3 - 25 - Softening Point

(Application for copies should be addressed to the American Oil Chemists Society, P.O. Box 3489, Champaign, IL 61821.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be a coal-based wax meeting the following requirements:

3.1.1 Physical and chemical properties. The desensitizing wax shall conform to the limits specified for the properties listed in Table I, when determined as specified in the applicable subparagraphs of 4.5. For commonly known Type I and Type II coal-based waxes see 6.1.1.

TABLE I
Physical and chemical properties of Type I and Type II coal-based waxes

Property	Requirement		Test Para.
	Type I	Type II	
Melting Pt. Min Deg F(deg C)Max	174 (79) 194 (90)	174 (79) 194 (90)	4.4.1
Viscosity at 210 deg F(98 deg C) (in Centistokes (cSt))	38+/-6	25+/-5	4.4.2
Penetration (in units of 0.1mm) 77 deg F/25 deg C Min 110 deg F/43 deg C Max	2.5 10	0.5 15	4.4.3
Flash Pt deg F (C) Min	400(204)	400(204)	4.4.4

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Organic Insoluble Matter % Max	0.03	0.03	4.4.5
Alkalinity, % Max	None	None	4.4.6
Acidity, as Acetic Acid, % Max	0.02	0.02	4.4.7
Oil Content % WT, Max	8%	12%	4.4.8
Exudation at 160 deg F (71 deg C), % Max	3.0	3.0	4.4.9
Congealing pt. Max	185(85)	181(83)	4.4.11
Deg F(deg C) Min	165(74)	165(74)	
Softening pt. Max	192(89)	187(86)	4.4.12
Deg F(deg C) Min	171(77)	172(78)	
FTIR for information only			4.4.13

3.2 First article testing. This specification makes provision for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract (see 4.2, 6.2 and 6.3).

3.3 Differential scanning calorimetry (DSC). - DSC curve(s) shall be obtained for the qualified product sample in accordance with 4.4.10 and utilized as a standard. DSC curves for future wax purchases shall be compared and purchased against this standard, and must exhibit major peaks/characteristics as shown in the standard.

3.4 Reactivity at 100 deg C or 120 deg C. The wax reactivity values shall not exceed 1 milliliter (ml), when determined as specified in 4.4.14. This test is to be performed by a Government laboratory (see 6.4).

3.5 Workmanship Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected. The wax lot shall be homogeneous through the lot, and of uniform texture /color, free from any foreign materials and contaminants.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as parts of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.2)
b. Quality Conformance Inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. Prior to initiation of regular production the contractor shall submit a first article sample consisting of 200 lbs of wax in accordance with instructions issued by the Contracting Officer for evaluation in accordance with 4.2.2. All samples submitted shall have been produced by the contractor using the same production process, procedures, and shall be accompanied by certificates of analysis. Supplemental first article samples, or portion thereof, as directed by the contracting officer, shall also be submitted in accordance with the contract provision: Instruction Regarding Submission of First Article.

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4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 or 4.4 of this specification and any or all requirements of the applicable drawings and specifications.

4.2.3 Re-examination. Supplemental first article inspections shall be required by the preparing activity under any of the following conditions:

(1) The manufacturer has modified the product or changed the material or processing sufficiently so that the physical and chemical properties of the product may be altered.

(2) The requirements in the specification have been revised sufficiently to affect the character of the product.

(3) When deemed necessary to determine that the product continues to meet all of the specification requirements.

(4) A major change in the product has occurred (see 6.3.2).

4.2.4 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspections upon any failure of a sample to comply with any of the stated requirements.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. A lot shall consist of that quantity of desensitizing wax produced by one manufacturer, in accordance with the same specification or same specification revision under one continuous set of operating conditions. The lot size shall not exceed 70,000 lbs of material that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. Inspection lots shall comply with the lot formation provision of MIL-A-48078 and MIL-STD-1168.

4.3.2 Examination and tests See MIL-A-48078.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a

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minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in ccordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Table II
Attributes sampling inspection

Lot Size	Inspection levels I	II
2 to 8	*	5
9 to 15	*	5
16 to 25	*	5
26 to 50	32	5
51 to 90	32	13
91 to 150	32	13
151 to 280	32	20
281 to 500	32	20
501 to 1200	80	20
1201 to 3200	80	32
3201 to 10000	125	32
10001 to 35000	125	50
35001 to 150000	125	50
150001 to 500000	200	50
500001 and over	200	50

Numbers under inspection levels indicate sample sizes; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides

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equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1	Waxes, Coal-Based		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Melting point	4.3 3	3.1.1	4.4.1
102	Viscosity	4.3.3	3.1.1	4.4.2
103	Penetration	4.3.3	3.1.1	4.4.3
104	Flash point		3.1.1	4.4.4
105	Organic insoluble matter	4.3.3	3.1.1	4.4.5
106	Alkalinity	4.3.3	3.1.1	4.4.6
107	Acidity	4.3.3	3.1.1	4.4.7
108	Oil content	4.3.3	3.1.1	4.4.8
109	Exudation	4.3.3	3.1.1	4.4.9
110	Congealing point	4.3.3	3.1.1	4.4.11
111	Softening point	4.3.3	3.1.1	4.4.12
112	DSC	4.3.3	3.3	4.4.10
113	Reactivity	4.3.3	3.4	4.4.14
Minor 201	Evidence of poor workmanship	4.3.3	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.2	Container, Sealed	1 OF 1	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged, exposing wax contents	Level I	5.1 and 5.2	Visual
Minor				
201	Marking misleading or unidentifiable	Level II	5.3	Visual

NOTES:

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4.3.3 Sampling for test. A representative sample of approximately 5 pounds of the wax shall be selected from each lot of coal-base wax submitted for acceptance. The sample shall be tested for the chemical and physical property requirements of Section 3 of this specification and the results recorded (see 6.2.2). Failure of the sample to meet the applicable requirements of 3.1.1, the lot shall be rejected.

4.4 Methods of inspection (see 6.6). The tests in 4.4.1 through 4.4.14 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.4.1 Melting point. The melting point shall be determined in accordance with ASTM Method D127. If the sample fails to meet the melting point requirement of 3.1.1, the lot shall be rejected.

4.4.2 Viscosity. The viscosity shall be determined in accordance with ASTM Method D445 using the Ubbelohde viscometer. If the sample fails to meet the viscosity requirement of 3.1.1, the lot shall be rejected.

4.4.3 Penetration. The penetration shall be determined in accordance with ASTM Method D1321. If the sample fails to meet the penetration requirement of 3.1.1, the lot shall be rejected.

4.4.4 Flash point. The flash point shall be determined in accordance with ASTM Method D92. Failure of the sample to meet the flash point requirement of 3.1.1 constitutes a cause for lot rejection.

4.4.5 Organic insoluble matter. The percentage of organic insoluble matter shall be determined in accordance with the following procedure. Failure of the sample to meet the organic insoluble matter requirement of 3.1.1 shall constitute a cause for lot rejection.

The solvent for this determination shall be aliphatic naphtha. Use clean medium porosity glass filtering crucibles which will allow 20 ml of hot solvent to flow through them, under suction, within 15 seconds or less. Condition the crucibles prior to use as follows: wash three (3) 20 ml portions of hot solvent through the crucible with the aid of suction, wash one (1) 20 ml portion of acetone through the crucible, back wash the crucible with one (1) 20 ml portion of acetone, dry the crucible in an oven at 100+/-5 deg. C, cool the crucible in a desiccator,

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weigh the crucible and record weight, return crucible to oven and maintain at 100+/-5 deg C until used.

Weigh approximately 5 grams of the sample to the nearest 0.1 milligram and transfer to a 400 ml beaker. Add approximately 250 ml of solvents to the beaker. Dissolve the wax in the solvent by placing the beaker and contents on a steam bath and heat at a temperature of 65 deg. C minimum. Continue heating until the wax dissolves (stirring may be used). Filter the hot solution through the previously conditioned and tared hot crucible with the aid of suction. Maintain the crucible and contents at above 60 deg. C to prevent the wax from coming out of the solution. Use a heat gun or other commercially available equipment to maintain the temperature of the solution. Wash the crucible with hot solvent until the residue and crucible are free of wax, or until approximately 5 ml of the washings, when evaporated to dryness, leaves no residue.

Retain the filtrate for the determination of alkalinity as specified in 4.4.6. During filtration air must not be sucked through the crucible. This is to prevent the air from cooling the filtering surface which could cause the wax to come out of solution and clog the crucible. Dry crucible and contents in an oven for 30 minutes at 100 deg. C to 105 deg. C, cool in a desiccator and weigh. The percentage of insoluble matter is calculated as follows:

Percent insoluble matter = 100A/W

Where:

A = increase in weight of filtering crucible, in gm.

W = Weight of sample, in gm.

4.4.6 Alkalinity. The sample shall be tested for alkalinity in accordance with the following procedure. Failure of the sample to meet the alkalinity requirement of 3.1.1 shall constitute a cause for lot rejection.

The filtrate from the determination of insoluble matter (4.4.5) shall be transferred to a 500 ml separatory funnel, and the flask shall be rinsed with two, 10 ml portions of aliphatic naphtha. Twenty-five ml of distilled water shall be added to the funnel. The funnel shall be shaken, the phases allowed to separate, and the aqueous layer withdrawn. The procedure of adding distilled water, shaking, and separating shall be repeated two more times. The aqueous extracts shall be combined. The extract shall be tested with red litmus paper of such sensitivity that its color is completely changed in one minute by .004 Normal

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(N) sodium hydroxide solution. The water extract shall be considered to be alkaline if upon application red litmus paper changes color within 30 seconds. The water extract shall be reserved for determination of acidity (see 4.4.7).

4.4.7 Acidity. The sample shall be tested for acidity in accordance with the following procedure. Failure of the lot to meet the acidity requirement of 3.1.1 shall constitute a cause for lot rejection.

Three drops of phenolphthalein indicator shall be added to the water extract, obtained as specified in 4.4.6, and the solution shall be titrated with approximately 0.1 N sodium hydroxide solution. A blank determination shall be made on the same quantities of aliphatic naphtha and water as were used in dissolving and extracting the sample. Any acidity shall be calculated to percentage of acetic acid in the sample as follows:

$$\text{Percent acetic acid} = \frac{6.0\ (A-B) \times N}{W}$$

Where

A= ml of sodium hydroxide solution used in determination
B= ml of sodium hydroxide solution used in blank
N = normality of sodium hydroxide solution
W = weight of sample, in gm.

4.4.8 Oil content. The oil content shall be determined in accordance with ASTM Method D721. If the sample fails to meet the oil content requirement of 3.1.1, the lot represented by the sample shall be rejected.

4.4.9 Exudation. The sample shall be tested for exudation in accordance with the following procedure. Failure of the sample to meet the exudation requirement of 3.1.1 shall constitute a cause for lot rejection. Vendor certification of performance and results of this test is acceptable.

The sample shall be melted at approximately 220 deg. F. When completely molten, the wax shall be cooled with stirring, to about 10 deg. F above its drop melting point and cast into a glass mold, 1 inch inside diameter and 0.5 inch high. The glass mold shall rest on a smooth 0.5 inch thick aluminum plate and at the time of casting, both plate and mold shall be at 70 deg. F +/- 10 deg. F. Under these conditions the sample will freeze in 3 to 5 minutes. The sample shall be removed from the mold and placed on 20 layers of filter paper, cut in 1 1/2 inch diameter circles. An 80 gm brass weight, 1 inch in diameter, shall be

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placed on top of the wax casting. This assembled sample shall then be placed on a wire screen and stored in an oven for 24 hours at 160 +/- 1/2 deg F. Before assembly, the brass weight shall be weighed and then reweighed with the wax casting. Afr hot storage, wax and brass weight shall be removed from the filter paper while still warm, no attempt being made to separate brass weight and wax, and allowed to cool upside down (the wax resting on the brass weight). When cool, the wax and brass weight shall be weighed together. The difference in weights shall be calculated as percent lost during storage. The test shall be run in triplicate and the average of the three values shall be reported as the percent exudation.

$$\text{Percent exudation} = \frac{A-C}{A-B} \times 100$$

Where:

A = weight of wax plus brass weight before exudation test, in gm.

B = weight of brass weight, in gm

C = weight of wax plus brass weight after exudation test, in gm.

4.4.10. DSC procedure. The sample shall be tested with differential scanning calorimetry in accordance with the following procedure. Failure of the sample to meet the DSC requirement 3.3 shall constitute a cause for lot rejection.

Use a Perkin-Elmer, Model DSC-1B, DSC-2, DSC P-3-14, or equivalent instrument with the following conditions:

a. Sample size - 5-8 mg of compacted or resolidified wax.
b. Heating rate - 5-10 deg. C/Min.
c. Temperature range - 25-100 deg. C
d. Inert gas flow - 10-20 ml of gas/minute.
e. Chart speed - 20 mm/minute (approximate)
f. Sample conditioning - Heat sample to 5 deg. C above its endotherm and allow it to cool to 25 deg. C without changing its position in the DSC heating chamber. Reprogram sample under conditions a through e.

4.4.11 Congealing point A congealing point will be obtained for each wax sample in accordance with the procedure specified in ASTM D938. Failure of the sample to meet the congealing point requirement in 3.1.1 shall constitute a cause for lot rejection.

4.4.12 Softening point. A softening point will be obtained for each wax sample in accordance with the procedure specified in

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A.O.C.S. Method Cc3-25. Failure of the sample to meet the softening point requirement in 3.1.1 shall constitute a cause for lot rejection.

4.4.13 FTIR. A FTIR (Fourrier Transform Infrared) will be obtained for the wax samples for information purpose using a Beckman Model FT1100 utilizing instrument operating procedure for the Model FT1100 or equivalent FTIR equipment may be used.

4.4.14 Determination of reactivity. The reactivity shall be determined in accordance with Method 504.1.1 of MIL-STD-650 with the following exceptions. Failure of sample to meet the reactivity requirement of 3.4 shall constitute a cause for lot rejection.

The specimens shall consist of 7.50 gm of wax, 5.00 gm of RDX and 5.00 gm of TNT. These specimens are reduced to a practicable and standardized fineness for intimacy of conduct and reproducibility of results, approximately 12 mesh.

Two and one half gm of wax shall be placed in each of three heating tubes. Two and one half gm of RDX shall be placed in one of the tubes containing wax and two and one half grams of TNT shall be placed in another tube containing wax. Two and one half gms of RDX and two and one half grams of TNT shall be placed in two additional tubes.

The sum of the volume of gases evolved by the wax and explosive individually subtracted from the volume of gas produced by the wax-explosive mixture shall be reactivity values to be reported.

5. PACKAGING

5.1 Packaging. Unless otherwise specified, commercial packaging in accordance with ASTM-D-3951 will be acceptable.

5.2 Packing. The wax shall be prepared for shipment in such a manner as to permit acceptance by carrier for transportation at the lowest applicable rate and to afford protection from normal hazards of transportation.

5.3 Marking. Marking of containers shall be in accordance with ASTM-D-3951 and such additional code markings, indicating stock number and ultimate destination, as may be specified by the procuring agency.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The waxes covered by this specification are intended for use as desensitizers in R8150 explosive.

6.1.1 Coal-based waxes. Coal-based waxes commonly known as montan waxes: Type I - Hoechst Celanese Montan KP wax or equal, Type II - Hoechst Celanese Montan S wax or equal. Hoechst Celanese Montan KP wax and Hoechst Celanese Montan S wax have been found satisfactory coal-based waxes for use in R8150 explosive. If equivalent substitute brands are to be used, first article test report(s) shall be forwarded for approval to the preparing activity (see 6.5) prior to initiation of full scale production.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)..
c. Provisions for submission/resubmission of first article samples.
d. Acceptance and description sheets - Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Name and address of manufacturer

6.2.2 Contract data requirements. For Army procurement, the data specified in paragraph 4.3.3 shall consist of the physical and chemical properties of the wax. A copy of all test results should be forwarded to Commander, ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, N.J. 07806-5000 (Data Item DI-R-1724 or equivalent).

6.3 Submission of first article samples

6.3.1 Instructions as to the location for evaluation of the first article sample shall be obtained from the contracting

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officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, ARDEC, AMCCOM, ATTN: SMCAR-QAR-R, Picatinny Arsenal, New Jersey 07806-5000. First article samples shall be identified by tags marked with the following information:

a. Wax, coal-based, Specification MIL-W-71094(AR)
b. Sample for First Article Tests
c. Manufacturers Formula Number or Brand Name
d. Date of shipment
e. Name and address of manufacturer

6.3.2 Definition of major changes. A major change in the product requires resubmission and retesting of first article samples (see 4.2.3). The determination of a major change in a manufacturer's product will be based upon the evaluation of the DSC curve for the following: Changes in characteristics from the original curve and subsequent curves for the same product in regard to shifts of the melting point or changes in the composition of wax (SINGLE CUT vs a blend) or melting point characteristics as demonstrated by the appearance of new peaks or disappearance of peaks in the DSC curve. The determination of a major change in the manufacturer's product for congealing and softening points will be based on a shift of greater than 5.0 deg. C (9.0 deg. F) from original values. Waxes produced in a location other than that specified plant, even though the same blend stocks and the same manufacture process are used, will be considered a major change of the product.

6.4 Reactivity. This test is applicable to the first article testing unless otherwise specified by the procuring activity. The temperature at which the test is to be run should also be specified by the procuring activity.

6.5 Preparing activity. Picatinny Arsenal is the preparing activity for this specification and is responsible for the coordination of specification testing. The address for the preparing activity is: Commander, ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, N.J. 07806-5000.

6.6 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ATTN: SMCAR-QAR-R, ARDEC, Picatinny Arsenal, N.J. 7806-5000. This description should include but not be limited to the procedures

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used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.7 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions shall be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.8 Subject term (key word) listing.

Montan waxes
Coal-based waxes
RDX
Explosive desensitizing waxes

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 9160-A078)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: MIL-W-71094(AR)	2. DOCUMENT DATE (YYMMDD): 920624

3. DOCUMENT TITLE: WAXES, COAL-BASED, FOR USE IN R8150 EXPLOSIVE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial: (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME: US ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6671 (2) AUTOVON: 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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